北京郵電大學

Queen Mary University of London

For examiners' use only

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Total	

EBU5376 A

Joint Programme Examinations 2023/24

EBU5376 Digital Signal Processing

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

Allowed: Basic calculators.
NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Yixuan Zou, Mr Andy Watson, Dr Richard Clegg.

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) **Figure 1** shows the block diagram of a general DSP system. **[4 marks]**

Briefly explain the function of

i) the Analog Filter **(2 marks)**

ii) the ADC **(1 mark)**

iii) a Zero-order Hold (ZOH) **(1 mark)**

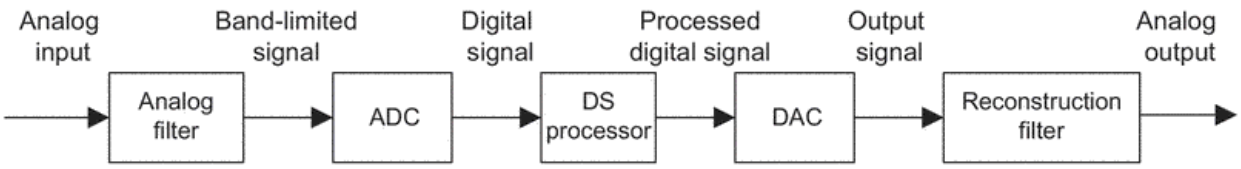

Figure 1

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	4 marks

b) This question is about poles and zeroes. **[4 marks]**

A particular function of Z is:

$$X(Z) = \frac{Z^2 - 2Z - 15}{Z^2 + 4Z + 12}$$

Determine zeroes and poles of the function $X(Z)$.

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	4 marks

c) This question is about deciBels (dBs). **[4 marks]**

i) What amplitude ratio corresponds to a dB value of -6dB? **(2 marks)**

ii) dBm is a way of referencing a power value in Watts to a power of 1 mW. A particular power value is quoted as 40dBm. Calculate the value of this power in Watts. **(2 marks)**

	Do not write in this column
	4 marks

d) This question is about sampling and quantisation. **[9 marks]**

Consider the sine wave given by $x(t) = sin(12\pi t)$.

i) What is the signal's frequency? **(1 mark)**

ii) Calculate (to 3 decimal places) the first 4 samples of $x(t)$ if the sampling period is 0.3 seconds.

(2 marks)

iii) State the quantised values if the quantisation levels are (-0.5, -1.0, 0, 0.5, 1.0). **(2 marks)**

iv) What is the sampling frequency? **(1 mark)**

v) Referring to Nyquist's Sampling Theory, comment on the choice of sampling frequency for this signal. **(3 marks)**

	Do not write in this column
	9 marks

~~e) This question is about convolution.~~ **~~[4 marks]~~**

~~$x[n]$ and $h[n]$ are two periodic sequences. Given that $x[n] = [1, -1, 2, 3]$ and $h[n] = [2, 4, -2, 1]$,~~

~~determine the convolution of $x[n]$ and $h[n]$ using the matrix method.~~

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	4 marks

Question marking: $\frac{}{4} + \frac{}{4} + \frac{}{4} + \frac{}{9} + \frac{}{4} = \frac{}{25}$

Question 2

a) This question is about the z-transform.. **[8 marks]**

i) State the equation of the z-transform $X(z) = Z(x[n])$ of a sequence $x[n]$. **(4 marks)**

ii) Give the z-transform of the sequence $x[n]$ such that $x[n] = 1$ if $-3 < n < 3$ and $x[n] = 0$ otherwise. **(4 marks)**

	Do not write in this column
	8 marks

b) This question is about the Fourier Transform. **[7 marks]**

i) Give the exponential formula for the Fourier Transform **(3 marks)**

ii) Given a function $x(t)$ with Fourier Transform $X(\omega)$. Show that the frequency shifted function $e^{j\omega_0 t}x(t)$ has the Fourier Transform $X(\omega - \omega_0)$ **(4 marks)**

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	7 marks

c) This question is about the Fast Fourier Transform (FFT).
Using Decimation-in-Time (DIT), plot the butterfly diagram and compute the FFT of the following 4-point sequence:

$$x[n] = [1, i, 2i, -1]$$

[10 marks]

For 2024-25, you are not required to memorize the butterfly diagram. The diagram will always be given in the question

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	10 marks

Question marking: $\frac{}{8} + \frac{}{7} + \frac{}{10} = \frac{}{25}$

Question 3

This question is about the basic representations of filters.

a) There are two types of digital filters, namely, Infinite Impulse Response (IIR) and Finite Impulse Response (FIR). Given a filter with the following difference equation

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2].$$

For what values of a_1, a_2, b_0, b_1, b_2, is this a FIR filter? Briefly explain your answers.

[3 marks]

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	3 marks

b) A particular digital filter has input and output variables $x[n]$ and $y[n]$ related by the difference equation:

$$y[n] + 2y[n-2] = x[n] + 0.5x[n-2]$$

[15 marks]

i) Draw the filter's signal flow graph. **(4 marks)**

ii) Determine the poles and the zeros of this filter. **(7 marks)**

iii) Draw the pole-zero plot of this filter. Based on the plot, discuss the stability of the filter.

(4 marks)

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	15 marks

c) The functionality of a filter can be examined by looking at various time-domain responses and frequency responses.

[7 marks]

i) Suppose I want to understand how a filter reacts to sudden changes in the signal, which filter response should I investigate? Is this a time-domain response or a frequency domain response?

(3 marks)

ii) If you are given the difference equation of the filter, how to determine the response that you selected in part (i)? Briefly describe your approach.

(4 marks)

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	7 marks

Question marking: $\frac{}{3} + \frac{}{15} + \frac{}{7} = \frac{}{25}$

Question 4

This question is about the filter design.

a) The cut-off frequency of a digital filter is usually represented on a normalized scale. **[6 marks]**

i) Explain the meaning of a normalized frequency. **(2 marks)**

ii) Suppose a low-pass digital filter has a normalized cut-off frequency of 0.2 radians/sample under a sampling rate of 10Hz. Explain how does this filter process a 10Hz sinusoid. **(4 marks)**

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	6 marks

b) Butterworth filter is a commonly used filter in digital signal processing. **[8 marks]**

i) Discuss the main advantage of using a low-pass Butterworth filter for digital signal processing. **(3 marks)**

ii) With the help of a diagram, describe the characteristics of the amplitude response of an ideal low-pass filter. **(5 marks)**

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	8 marks

c) You are asked to construct a digital Butterworth filter with a cut-off frequency of 0.2 rad/s under a sampling rate of 10Hz. It is given that, under a sampling interval of T seconds, the relationship between analog frequency Ω and digital frequency ω follows $\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$. Also, the equation of Bilinear Transform is

$$H(z) = H(s)\big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

where $H(z)$ and $H(s)$ denote the transfer functions of the digital and analog filters, respectively. T denotes the sampling interval. Based on the given information, answer the following questions

[11 marks]

i) To construct a digital Butterworth filter, you need to first choose the appropriate analog Butterworth prototype. How are you going to determine the cut-off frequency of the analog prototype? **(5 marks)**

ii) You are given a Butterworth prototype with the following transfer function:

$$H(s) = \frac{1}{\frac{s}{\omega_c} + 1},$$

where ω_c denotes the cut-off frequency. Using Bilinear Transform, obtain the difference equation of the digital filter that satisfies the requirements specified in the beginning.

(6 marks)

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	11 marks

Question marking: $\frac{\quad}{6} + \frac{\quad}{8} + \frac{\quad}{11} = \frac{\quad}{25}$

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